

POWER SECTOR DEVELOPMENT IN CENTRAL ASIA.

FEBRUARY 2025

CHALLENGES AND ROAD AHEAD



CONSULTING
ENGINEERS



UNLOCKING CENTRAL ASIA'S POWER SECTOR

Population growth, industrialization and reform, along with vast fossil fuel reserves and renewable energy potential offer significant development opportunities in Central Asia's power sector. Challenges such as aging infrastructure, low tariffs and regulatory uncertainty remain and call for policy changes, greater public-private collaboration and improved regional integration to unlock the sector's potential.

REGIONAL ENERGY LANDSCAPE

Driven by rapid economic growth, infrastructure expansion and modernization of industry and energy systems, Central Asia is undergoing a dynamic phase of development, with a long-term focus on energy security, sustainability and regional integration.

Kazakhstan, Uzbekistan and Turkmenistan are endowed with vast fossil fuel resources including coal, oil, and natural gas. With their mountainous rivers, Tajikistan and the Kyrgyz Republic have enormous hydropower reserves. Kazakhstan and Uzbekistan also possess significant untapped solar, wind and hydropower potential. This distribution of resource wealth offers the region a dual pathway to sustainable development and energy security.

These countries are seeking to unlock their energy potential for different strategic reasons. Faced with deficits in energy generation, Kazakhstan and Uzbekistan strive to expand and diversify their energy sources to meet the needs of their growing economies and populations, and to support their decarbonization commitments. Due mainly to seasonal fluctuations in energy demand, a key priority for the Kyrgyz Republic and Tajikistan is to address energy security concerns, especially during the winter months. Resource-rich Turkmenistan (natural gas and hydropower), Tajikistan and the Kyrgyz Republic (both hydropower) are looking to expand their export potential – both within and outside Central Asia.

While Kazakhstan, Uzbekistan and Turkmenistan are investing heavily in domestic energy infrastructure, energy mix diversification and reduced reliance on imports, there is also growing recognition of the importance of more effective regional cooperation. Improved collaboration, in the context of the Central Asian Power System (CAPS), among others, is essential for enhancing grid interconnectivity and the stability of national energy systems – especially when considering the growing demand for energy, the need for energy security, and the transition to more sustainable and renewable energy sources.

CHALLENGES AND OPPORTUNITIES

With generation facilities and electricity grids dating back to the Soviet era, a pressing concern across the region is that of aging infrastructure. High transmission and distribution losses are prevalent, with regional blackouts becoming more frequent. To accommodate the integration of large-scale variable renewable energy capacities and ensure the stability and reliability of energy systems, the region's energy infrastructure – especially power lines – urgently requires modernization.

Attracting large-scale foreign investment and financing for infrastructure modernization, clean energy development and energy efficiency improvement is a serious challenge, the success of which depends primarily on the creation of robust incentives for potential investors, and the development of comprehensive legal and regulatory frameworks. In this respect, Central Asian countries have made varied progress. Kazakhstan and Uzbekistan have shown greater regulatory readiness for decarbonization, with specific targets and policies in place to support renewable energy development. The Kyrgyz Republic and Tajikistan, with lower carbon footprints, focus more on hydropower development, but lack appropriate mechanisms to attract private investors.

Nevertheless, Central Asia offers numerous opportunities for a diverse range of stakeholders, with incentives to help unlock the region's energy potential. The region's endowment in fossil fuels will continue to serve as a foundation for sustained energy exports, while its abundant renewable energy resources can drive a cleaner energy transition, both within the region and beyond, with a long-term focus on export potential. International support from multilateral development banks, such as the World Bank, Asian Development Bank, and European Bank for Reconstruction and Development has already strengthened the green transition through financing and technical assistance.

REGIONAL POWER SECTOR CONTEXT

Central Asian countries are shifting their focus back to strengthening regional energy cooperation by reintegrating power systems, increasing electricity trade, and developing infrastructure. With plans for a regional electricity market and increased integration of renewable energy, Central Asia is paving the way for greater reliability of power systems and broader international cooperation in the future.

WHAT IS CAPS?

Also known as the “Energy Ring,” the Central Asian Power System (CAPS) is a regional electricity grid, historically providing an interconnection between the power grids of Uzbekistan, the Kyrgyz Republic, Turkmenistan, Tajikistan, and the southern region of Kazakhstan. Its dispatch center is located in Tashkent, coordinating power distribution across the region.



HISTORICAL BACKGROUND

Dating back to the Soviet era, CAPS was fully established by the 1970s to optimize energy distribution across Central Asia by balancing seasonal fluctuations in supply and demand, fostering regional cooperation, and enhancing overall grid reliability. However, it operated independently of the USSR’s Unified Power System, limiting broader integration.

The system allowed for more efficient use of available resources in a cost-effective manner, ensuring energy security across the region. Hydropower plants in the Kyrgyz Republic and Tajikistan supplemented the thermal energy produced by Uzbekistan, Kazakhstan and Turkmenistan. This complementary energy mix made use of both renewable and fossil-fueled generation.

During the winter, upstream countries stored water while receiving electricity from the thermal power stations of downstream countries. During the agricultural season, water was released for downstream irrigation, thereby generating electricity. In this context, CAPS also played a critical role in regional water resource management, balancing energy needs and ensuring agricultural productivity in arid regions.

However, with the dissolution of the USSR, and these nations developing their own independent

national energy priorities, this cross-border interconnection became underutilized. Supported by its abundant hydrocarbon resources, Turkmenistan left CAPS in 2003. Kazakhstan withdrew from CAPS in 2008, followed by Uzbekistan in 2009; Tajikistan has also been essentially disconnected since 2009, further fragmenting the once efficient and unified system.

However, three countries – Kazakhstan, Uzbekistan and the Kyrgyz Republic – remained in some form of cooperation, with their energy systems operating in parallel.

The isolationist approach with underutilization of cross-border interconnection exposed vulnerabilities of national grids, leading to numerous disruptions in power supply, with blackouts in major cities, as well as shortages during winter.

Only in recent years, seeking to address these issues and due to rapidly increasing energy demands, have the countries shown a renewed interest in reestablishing CAPS. Financed by the Asian Development Bank, Tajikistan is in the process of rejoining CAPS by interconnecting with Uzbekistan’s power system. This marks a significant step toward restoring the regional interconnectivity and promoting further future collaboration on cross-border electricity trade.

Coal



KAZAKHSTAN

Hydro



KYRGYZSTAN

Natural Gas



UZBEKISTAN

Hydro



TAJIKISTAN

Funded by the Asian Development Bank, Tajikistan will reconnect to CAPS through interconnection with Uzbekistan

Natural Gas



TURKMENISTAN

Turkmenistan has disconnected from CAPS to operate in parallel with Iran

GENERATION CAPACITIES

The energy systems of these countries are shaped by their distinct natural resources, resulting in a diverse energy mix across the region.

Kazakhstan relies heavily on fossil fuels (primarily coal), which accounts for 70-75% of its electricity generation.

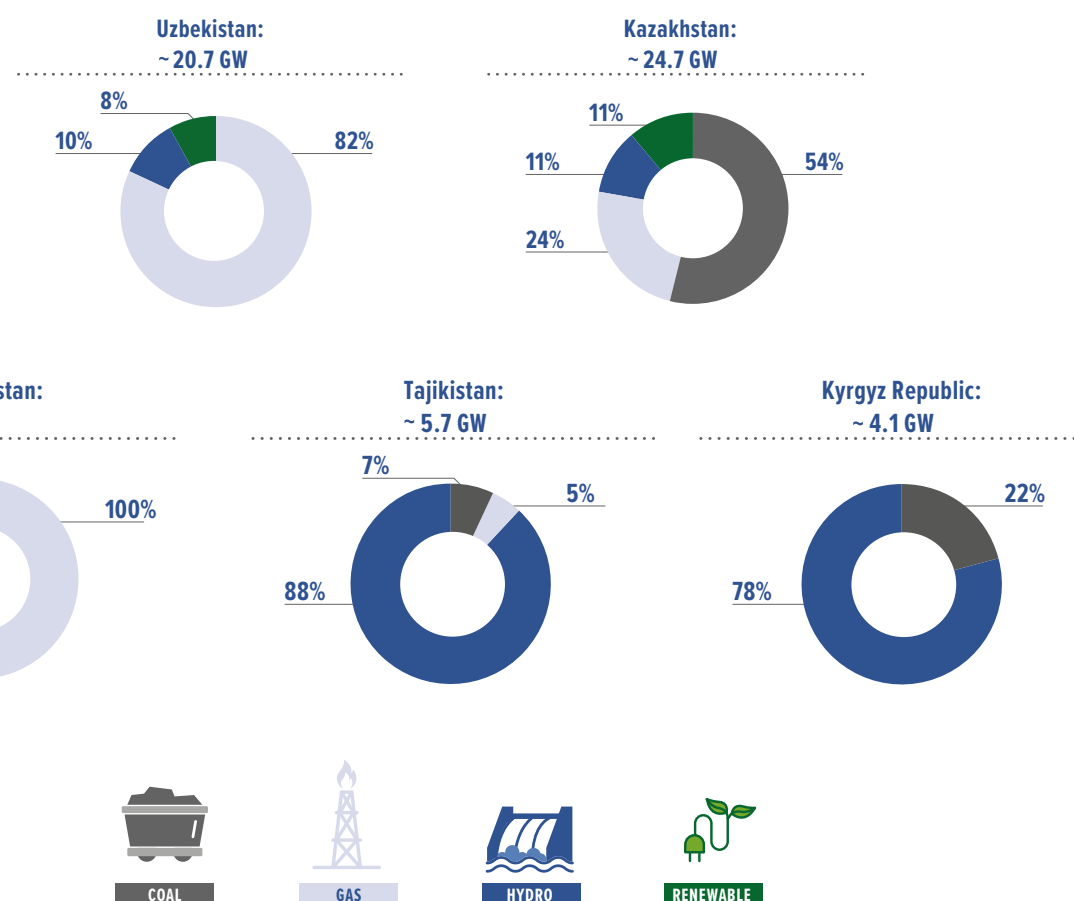
Uzbekistan's system is dominated by natural gas, which accounts for 85-90% of its power generation. The **Kyrgyz Republic** and **Tajikistan**, the region's upstream countries, rely predominantly on hydropower, with over 90% of electricity coming from water resources.

Turkmenistan, with abundant natural gas, generates nearly all of its electricity in gas-fired power plants.

Both Kazakhstan and Uzbekistan have already started to diversify their energy mix with renewables, primarily wind and solar power.

Over the last ten years, the total installed capacity of renewables in Kazakhstan has increased from 178 MW in 2014 to over 3 GW in 2024. The total installed renewable energy capacity (solar and wind) in Uzbekistan is estimated at 1.6 GW, with ambitious plans to reach approximately 20 GW by 2030.

Nevertheless, the region's overall energy infrastructure remains reliant on fossil fuels and hydropower, as renewable energy sources, while growing steadily, still account for a relatively small share of total energy mix.



CROSS-BORDER INTERCONNECTION UTILIZATION

Currently, the power systems of Kazakhstan (its southern region), the Kyrgyz Republic and Uzbekistan operate in parallel within the framework of CAPS, connected by 220-500 kV transmission lines. Recently, Tajikistan's power system initiated reconnection to CAPS through new 500 kV transmission lines, connecting it to Uzbekistan's grid.

With the current Rogun hydropower project to be completed by 2035, with an expected installed capacity of 3 600 MW, this return is a promising step towards a more interconnected and reliable regional grid. Turkmenistan, although formally disconnected from CAPS, continues to cooperate with the region by exporting electricity to Uzbekistan and the Kyrgyz Republic using stand-alone generators, connected to Uzbekistan's power grid.

Electricity supply agreements are bilateral in nature, except for transit purposes. For example, Uzbekistan and Tajikistan have annual agreements under the water-energy nexus. Similarly, Kazakhstan supplies the Kyrgyz Republic with electricity to support water management efforts.

In 2023, Central Asian countries actively traded electricity, both within CAPS and with external players, including Afghanistan (as an importer) and

Russia (as an exporter). Uzbekistan, Kazakhstan and the Kyrgyz Republic were net importers of electricity due to rapidly growing demand. By contrast, Tajikistan and Turkmenistan remained net exporters.

Further integrated development of power systems in the region depends on the establishment of a regional electricity market, an initiative that is still in its early stages. If realized, it could expand cooperation beyond Central Asia – to Azerbaijan, Turkey, Iran, Pakistan and Afghanistan. Mega-projects, such as CASA-1000, aiming to deliver Central Asian electricity to South Asia, also reflect the region's commitment to fostering cooperation with external partners.

Another integrative process is underway to the north – with efforts to create a single electricity market for multilateral cooperation under the EAEU framework, and to connect the Russian grid to the CAPS infrastructure.

Changing the course towards further integration of power systems in Central Asia is crucial for enhancing regional energy security and facilitating the integration of renewable energy sources. This shall be supported by both establishing common mechanisms for trade and upgrading infrastructure.

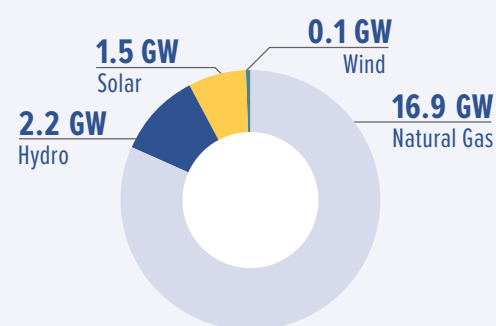
	(GWH)	IMPORT						TOTAL
		Uzbekistan	Kazakhstan	Kyrgyz Republic	Tajikistan	Turkmenistan	Russia	
EXPORT	Uzbekistan			147.7	18.2			1 989.4
	Kazakhstan			1 449.9			1 377.1	1 440.9
	Kyrgyz Repub.	82.1	68.2		2.6			152.9
	Tajikistan	907.5		145.0				2 587.0
	Turkmenistan	4 014.6		1 770.7				5 785.3
	Russia		4 994.4					4 994.4
TOTAL:		5 004.2	5 062.6	3 504.3	20.8	0.0	1 377.1	4 758.0

Table 1. Volumes of electricity imports & exports to & from CAPS countries in 2023



UZBEKISTAN

Uzbekistan’s power sector reform aims to bridge the electricity gap between supply and demand and transition to a green economy by 2030. Key initiatives include market liberalization and renewable energy expansion, although challenges remain, including low tariffs and outdated infrastructure. Strong political commitment and foreign investment create promising opportunities for growth and development.



GENERATION

- 17 thermal power plants
- 60 hydropower plants
- 9 renewable energy plants
- Available capacity: 20.7 GW

TRANSMISSION

- Grid operated by NEGU JSC
- 220-500 kV (10 700 km)
- 500 kV substations (7)
- 220 kV substations (70)

DISTRIBUTION

- Distribution network operated by Regional Electric Grids JSC
- 35-110 kV substations (1 805)
- 0.4/6/10 kV-110 kV (~279 000 km)

POWER SECTOR DEVELOPMENT

Driven by plans to address the widening gap between electricity supply and demand, and to move towards a green economy, reform in Uzbekistan’s power sector is underway, both in terms of its energy supply mix and its transition to a liberalized market.

Natural gas will remain the backbone of electricity supply, but to a lesser degree than at present, with the planned development of renewable energy capacity to meet Nationally Determined Contributions (NDC) targets (35% drop in greenhouse gas emissions by 2030, compared to 2010 levels). The transition to a fully liberalized market is expected to be completed by 2030.

Several factors have catalyzed sector reform, including strong political will, the role of transaction advisors, and procurement mechanisms (backed by investment laws) that have rapidly sped up foreign investment. A key factor has been a dramatic shift from tender-based to directly negotiated power purchase agreements.

Low end-user tariffs remain a key obstacle to sector reform. Another obstacle to development is the severely outdated grid infrastructure, with the need to upgrade the country’s transmission and distribution system seen as a key issue. The system will also require increased balancing capacity, as renewable energy becomes more prominent in the energy supply mix. With the planned shift away from gas-fired thermal plants, other large-scale technologies will become more relevant in the long term to balance a large share of volatile renewable energy in the supply mix.

MARKET DYNAMICS

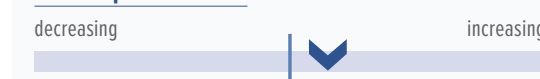
Uzbekistan’s power sector was historically characterized by centralized, public sector participation only, and a high degree of subsidization of electricity tariffs. With the establishment of the Ministry of Energy in 2019, the country began an ambitious process of sector-wide reform, including the transition to a competitive wholesale market, implementation of cost-reflective tariff policies and the appointment of an independent regulator.

Privatization is expected to increase in the medium term, primarily through increased development of renewable capacity through IPPs, as well as through a relatively small private contribution to hydropower projects.

Transmission is expected to remain closed to private investment; distribution is planned to include private sector participation, but a model for such participation has not been developed.

There has been considerable inertia in the transition to a liberal market, but the appointment of a regulator and the unbundling of NEGU in 2024 into separate operations and retail entities signals progress. Tariff subsidies are to be phased out by 2026, but the phase-out will not be without problems.

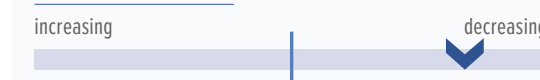
Market privatization



Market operation



Tariff subsidization



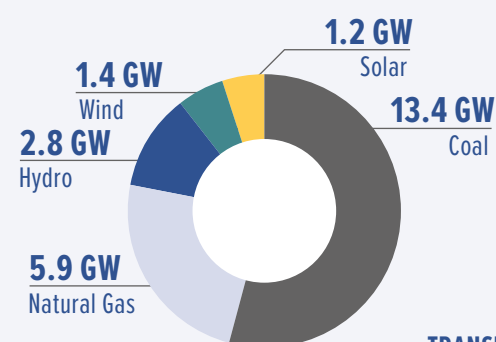
Outlook

Growing demand, strong political will to establish a renewables-heavy supply mix, abundant solar and wind resources, and fast-track procurement processes supported by a robust legal framework provide a solid case for future growth in the sector, with potential opportunities available to private investors. However, the grid’s ability to evacuate energy from high-potential areas along with low end-user tariffs remain key constraints to generation development, requiring serious consideration and action by policy-makers.



KAZAKHSTAN

Kazakhstan’s power sector is gradually transforming to rise to the dual challenge of meeting rapidly growing demand while pursuing decarbonization goals. Despite progress in market reforms and renewable energy initiatives, aging infrastructure, grid instability and regulatory challenges remain barriers to achieving long-term energy transition ambitions.



GENERATION

- 222 power plants
- 147 renewable energy plants
- Installed capacity: 24.7 GW
- Available capacity: 20.4 GW

TRANSMISSION

- National Electricity Grid operated by KEGOC JSC
- 220-1150 kV (27 234 km)
- 81 substations

DISTRIBUTION

- Distribution network operated by the Regional Electric Network companies (REC)
- 0.4-220 kV (>270 000 km)

POWER SECTOR DEVELOPMENT

Kazakhstan’s power market is evolving, driven by decarbonization targets and, more importantly, growing demand due to population growth and economic expansion. As part of the NDCs, the country aims to reduce greenhouse gas emissions by 15% economy-wide by 2030, compared to 1990 levels. The overarching goal for this carbon-intensive economy in the context of sustainable development is to achieve carbon neutrality by 2060. Meanwhile, projected supply and demand figures issued in 2024 show a growing deficit. With the current installed capacity of 24.6 GW, the system could be facing a shortfall of 6.24 GW generating capacity in 2030.

The country is committed to improving its energy infrastructure through investment in new generating capacities (including renewable energy through an auction mechanism), grid modernization and digitalization.

Despite these growth drivers, including strong political will to achieve sustainable development goals and continued support in establishing an enabling regulatory framework from IFIs, Kazakhstan faces several challenges that hinder the development of its electricity market. Grid stability is a major concern, along with aging infrastructure and overreliance on coal-fired power plants. Lack of institutional capacity in local executive agencies and a shortage of qualified personnel, as well as the absence of a well-represented independent regulatory body, all hinder the timely implementation of electricity market reforms.



Outlook

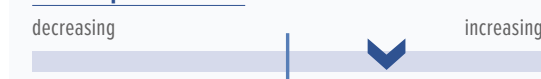
Kazakhstan’s abundance of renewable resources, especially its huge untapped wind power potential offers investors great opportunities. However, the country’s power sector is not without risks, such as a lack of technical feasibility studies and potential delays in grid expansion. The outlook is cautiously optimistic, as the government shows a strong commitment to modernization, with long-term success depending on the timely implementation of necessary reforms and infrastructure upgrades.

MARKET DYNAMICS

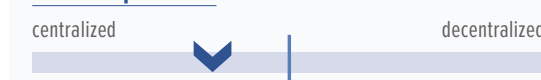
Since its independence in 1991, Kazakhstan has made structural reforms and has been an early adopter of a multi-market model, including decentralized (bilateral), centralized (spot for day-ahead and intra-day trading) balancing, ancillary services and capacity markets. This has allowed Kazakhstan to develop a relatively more efficient electricity market compared to the regional counterparts.

The recent establishment of the Single Buyer model and real-time balancing market aims to centralize market operations and optimize the operation of the energy system.

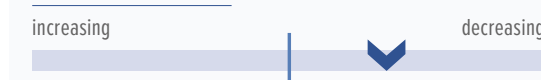
Market privatization



Market operation



Tariff subsidization



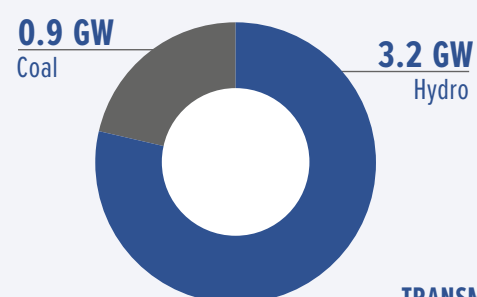
Private participation in the market is expected to increase with new IPPs, which could bring investments and technological advancements to the sector. However, more regulation is expected outside the generation sector, with transmission remaining state-owned and private retailers being excluded from the market. This limits full market liberalization, but maintains centralized control over critical infrastructure. There is a plan to phase out cross-subsidies by 2031, and tariffs are expected to increase gradually to reflect actual market costs, with no tariff differentiation for different consumer groups, and encourage energy efficiency.

The country has a clear policy focus on the transition to clean energy, with mechanisms in place to support the integration of renewable energy in to the grid: PPAs for 20 years at fixed tariffs / auction prices, annual indexation adjusted to the currency exchange rate, prioritized connection to the grid, land reservation during auctions, exemption from paying for transmission services and tax preferences.



THE KYRGYZ REPUBLIC

The Republic has declared a state of emergency in the power sector due to a supply-demand deficit caused by declining hydropower, population growth, and increasing industrialization. Solar and wind projects show potential, but outdated infrastructure, low tariffs, and centralized control hamper reforms. Another obstacle to development is the government's preference for public finance with respect to large hydropower projects.



GENERATION

- 7 large hydropower plants
- 27 small hydropower plants
- 2 combined heat & power plants
- Available capacity: 4.1 GW

TRANSMISSION

- Grid operated by NESK JSC
- 110-500 kV (~7 550 km)
- 110-500 kV substations (190)

DISTRIBUTION

- Falls under NEGK
- Length of T&D network (>80 000 km)
- Overall grid losses of circa 20%

POWER SECTOR DEVELOPMENT

In July 2023, the Kyrgyz Republic introduced a state of emergency in the energy sector until the end of 2026. Due to declining hydropower production (which accounted for almost 90% of the total electricity generated in 2023), population growth and increasing industrialization, the country is in the midst of a serious supply-demand deficit.

Bridging the deficit gap, particularly during the winter season, is of paramount importance to the country. With hydropower set to remain the key generator of electricity, but requiring long lead times for implementation, solar and wind technology can help to bridge the gap through relatively quicker implementation times.

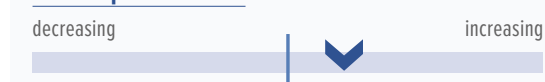
By introducing several new incentives for developers, the update to the Law on Renewable Energy Sources in August 2023 was a positive advancement in attracting investment in the sector. However, regulations supporting the law are not well-defined and highly unpredictable; it is unclear how the new incentives will be implemented in practice.

Historically, the sector has suffered from a lack of proper target setting and strategic planning. Moreover, a key impediment to private sector development is the government's preference for publicly-financed infrastructure projects. Severely outdated grid infrastructure also poses a major challenge by limiting the country's ability to evacuate wind and solar power. Tariffs, which are among the lowest in the region, are a major structural obstacle to sector reform.

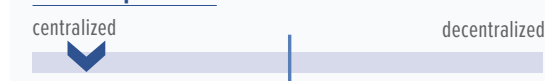
MARKET DYNAMICS

The Kyrgyz Republic's power sector is also characterized by structural volatility, as well as increased consolidation of power sector assets. A move towards liberalization took place in 2001, when the former state-owned energy company, Kyrgyzenergo, was split into several independent entities. These entities were reconsolidated in 2016, under the establishment of the National Energy Holdings Company (NEHC). Apart from private players in small hydropower, only four companies remain in the power sector.

Market privatization



Market operation



Tariff subsidization



Private sector participation is expected to increase marginally in the medium term, driven mainly by investments in solar and wind generation capacity. Large hydropower plants are unlikely to be privatized, as the government currently favors public funding of large-scale hydropower projects. No concrete plans are in place to liberalize the sector; therefore, market operation is expected to remain centralized.

By 2030, subsidization is expected to decrease, but persist, mainly due to the lack of an independent regulator and expected significant consumer resistance to tariff increases.



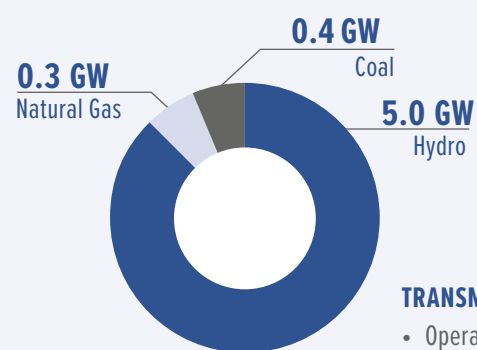
Outlook

Declining local production, population growth and increasing levels of industrialization have contributed to a major demand-supply deficit. As a consequence, a presidential decree has been issued, declaring a national state of emergency in the energy sector for the Kyrgyz Republic, thereby creating potential for future investment in capacity generation. Among other disabling factors, the government's preference for publicly funded large hydropower projects is a major barrier to private involvement in the sector.



TAJIKISTAN

Tajikistan's power sector will benefit from the Rogun Hydropower Plant's capacity to reduce supply shortages. The National Development Strategy aims for a capacity of 10 GW and more renewable sources by 2030. Challenges include aging infrastructure, subsidies and regulatory issues, but gradual reforms and private investment in renewables are expected.



TRANSMISSION

- Operated by JSC Shakakahoi Intiqoli Barq
- 220-500 kV (1 651 km)
- 10 substations (6 434 MVA)

GENERATION

- 14 hydropower plants & 3 thermal power plants
- 285 small hydropower plants
- 2 state-owned generation firms (Barki Tochik, Rogun HPP)
- 2 IPPs (Sangtudin HPP-1 & 2)
- GBAO concession
- Available capacity: 5.7 GW

DISTRIBUTION

- Operated by JSC Shabakahoi Tasqsimoti Barq
- 6-220 kV (> 27 000 km)
- 418 substations (9 225 MVA)

POWER SECTOR DEVELOPMENT

Resolving seasonal electricity supply shocks is a fundamental driver for development in a sector which is almost completely dependent on hydropower, and therefore susceptible to the impact of water level fluctuations, with major shortages occurring during winter. The Rogun HPP (3.6 GW when completed) will mitigate this by storing water in its reservoir for controlled energy production during low flow winters. The National Development Strategy (NDS) up to 2030 also envisions a transition towards greater industrialization for the country, with installed power capacity to increase to 10 GW, and annual electricity export to reach 10 TWh.

Supply diversification is another development driver, as the country plans to increase the share of non-traditional supply sources up to 10% by 2030. Decarbonization will be essential for meeting Tajikistan's carbon emissions reduction target under the Paris Agreement (30-40% by 2030, compared to 1990 base levels).

Aging infrastructure is a major constraint to sector development, with the NDS aiming to reduce power losses to 10% by 2030.

Tariffs are heavily subsidized, negatively impacting the state utility's finances and impeding investment in the sector. Regulatory complexity, lack of transparent procurement processes, and difficulties in securing land and permits also hinder private sector participation.

Sovereign risk, weak enforcement of contracts, and currency risk are other factors that deter international private investment.

MARKET DYNAMICS

The sector has been partially unbundled, with the energy monopoly OJSC Barqi Tojik having been unbundled into three separate entities. Barqi Tojik is still responsible for generation and market operations, with OJSC Shabakahoi Intiqoli Barq now responsible for transmission, and OJSC Shabakahoi Tasqsimoti Barq for distribution. Further unbundling would involve additional reforms, including decentralization of operations, appointing an independent regulator, and the development of a new tariff methodology that phases out subsidies.

Market privatization



Market operation



Tariff subsidization



Driven mainly by growth in small hydropower capacity and the government's diversification strategy into non-traditional sources of energy, a marginal increase in private sector participation is expected in the medium term. Market operation is expected to remain largely centralized, compounded by the absence of an independent regulator, as the Ministry of Economic Development and Trade continues to set electricity tariffs.

Subsidy reductions are likely to be gradual, with a focus on maintaining affordability for vulnerable populations while improving cost recovery for utilities.



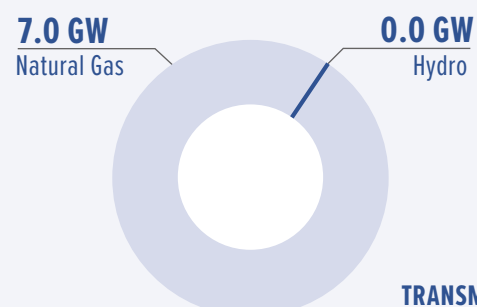
Outlook

Tajikistan's energy sector will evolve towards greater stability with the Rogun HPP mitigating seasonal supply shortages and supporting growth. While supply diversification and decarbonization efforts will boost the share of renewables, challenges such as aging infrastructure, tariff subsidies, and regulatory complexity will slow private sector involvement. Market operations will remain centralized, with incremental tariff reforms focused on improving cost recovery while maintaining affordability.



TURKMENISTAN

Turkmenistan is at a unique crossroad: abundant natural gas resources sustain its power generation and drive expanding electricity exports, yet the country is committed to reducing greenhouse gas emissions and improving efficiency. The country must balance its ambitions for renewable energy development with long-standing subsidy structures and infrastructure constraints on its path towards a more sustainable and interconnected energy market.



GENERATION

- 12 power plants owned by Turkmenenergo
- 1 hydropower plant (1.2 MW)
- Installed capacity: 7 GW

TRANSMISSION & DISTRIBUTION

- National Electricity Grid operated by Turkmenenergo
- 100% electrification, but inefficiencies in transmission system
- 35-500 kV, approx. 13 000 km

POWER SECTOR DEVELOPMENT

The power market is driven by plans to expand its electricity export potential, upgrade infrastructure, and, to a lesser extent, diversify its generation mix through the introduction of renewable energy technologies.

The energy system is dominated by natural gas, with the fourth-largest proven gas reserves globally. Electricity generation is almost entirely reliant on natural gas, with almost no alternative sources. Unlike other countries in the region, Turkmenistan does not face the issue of power deficit to cover domestic consumption. As a net exporter of electricity, Turkmenistan supplies power to Iran, Afghanistan, Uzbekistan, and the Kyrgyz Republic, steadily expanding both the volume and the reach of its exports. The construction of a new 1 574 MW combined-cycle power plant in the Balkan region, alongside grid infrastructure upgrades, shall enable the country to further extend its export routes to Afghanistan, South Asia, the Caucasus via Iran, as well as countries across Central Asia.

Turkmenistan has committed to a 20% reduction in CO₂ emissions by 2030, compared to 2010 levels. However, the country has consistently set set records for methane emissions, struggling to extinguish the burning Darvaza crater. Per capita greenhouse gas emissions remain above the global average. While significant efforts were made to enhance efficiency by introducing new power plants with reduced fuel consumption, e.g. a 1 500 MW combined-cycle power plant in Mary, the implementation of renewable energy technologies as a component of the green transition process is

still in its early stages. To date, only a few utility-scale renewable energy projects have gained momentum. Turkmenistan has substantial untapped renewable energy potential – some estimates put it at 655 GW for solar and at 10 GW for wind power.

Compared to the development of clean energy, much more emphasis has been placed on strengthening the national grid, including the construction of 1 100 km of new transmission lines (110-500 kV) and new substations, as well as the expansion of existing substations. This will provide Turkmenistan with a more reliable power grid.

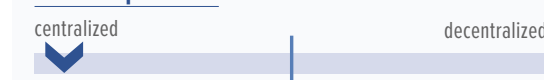
MARKET DYNAMICS

Since the 1990s, Turkmenistan's energy sector has been subsidized, providing free electricity to the citizens. Having abolished the free supply of electricity in 2019, the country has plans to move to a free market economy. However, the level of subsidization remains high, and electricity prices continue to be kept artificially low.

Market privatization



Market operation



Tariff subsidization



Outlook

While recent policy and legislation changes demonstrate the country's commitment to transitioning to cleaner power generation, it is still hampered by the existing level of energy subsidies and, therefore, the lack of adequate energy pricing mechanisms and special incentive schemes for investments in renewable energy and energy efficiency projects. To unlock the vast potential of both renewable energy and hydrogen, and move away from carbon intensive energy production, further action is needed to improve the investment climate.

Turkmenenergo, a state-owned, vertically integrated power company, continues to control the whole supply chain – from generation to transmission and distribution services. Structural reforms in the power sector have been slow, largely due to the government's continued reliance on central planning. The lack of market-based pricing mechanisms presents serious challenges for attracting foreign investment to improve the sector's efficiency.

UNLOCKING POTENTIAL

Growth, industrialization and a cleaner energy transition make for a promising development outlook for Central Asia's power sector. Countries such as Uzbekistan and Kazakhstan have clear strategic plans, and sector reform is underway. However, issues such as aging infrastructure, low tariffs and regulatory uncertainty continue to impede progress, calling for policy changes that could promote confidence and facilitate investment.

OPPORTUNITY LANDSCAPE

A promising outlook: Central Asia's power sector offers significant opportunities for development. Sector drivers, including widening demand-supply gaps due to economic expansion and growing industrialization, modernization of infrastructure, increased cross-border collaboration, diversification of the energy supply mix and policy shifts towards cleaner energy sources to meet decarbonization targets paint a positive picture for development in the region.

An enabling environment: several factors show promise in helping to create an enabling environment for this development. The region has abundant natural resources, including large hydrocarbon reserves, significant untapped solar and wind potential, and some of the largest hydropower resources in the world. Countries in the region have articulated clear development strategies (e.g., Uzbekistan, Kazakhstan and Tajikistan); these, in addition to decarbonization commitments under the Paris Agreement, signal strong political intent for both expansion and reformation. While not liberalized in general, certain countries in the region have made significant steps towards decentralization, private participation and, crucially, independent market regulation.

Stumbling blocks persist: aging infrastructure, particularly in transmission and distribution networks, remains an impediment to development. Other issues, such as low tariffs, regulatory uncertainty, lack of transparency, and institutional constraints continue to hinder progress. Moreover, political instability and economic uncertainty pose additional risks to development initiatives.

Strategic partnering to mitigate risk: engagement with local governments, multilateral organizations, and regional energy players will be crucial to navigating risks and successfully implementing projects. Forming joint ventures can help mitigate investment risks and enhance project feasibility.

POLICY RECOMMENDATIONS

Policymakers are encouraged to:

Develop comprehensive energy strategies that clearly articulate sector targets which are realistic, time-bound, and are in alignment with overarching national development goals.

Detail masterplans which outline clear, actionable steps for infrastructure development, integration of renewable energy and grid modernization, ensuring alignment with long-term sustainability goals, economic growth, and efficient resource utilization across sectors.

Decentralize energy sector operations and actively support increased private sector participation through enhanced financing conditions to boost efficiency, drive down costs, improve system reliability and resilience, and foster innovation.

Prioritize the establishment of clear, transparent, and consistent regulations that govern the power sector, particularly for renewable energy projects. A clearly defined and efficient procurement framework is a vital enabler to sector development. Moreover, crucial to an enabling regulatory environment is the appointment of an independent regulator.

Move towards tariff models that are transparent and cost-reflective, thereby ensuring financial sustainability and helping to attract private investment, while simultaneously adopting measures to protect vulnerable consumer bases, to balance long-term sector viability with tariff affordability.

Prioritize strengthening institutional capacity by improving organizational efficiency and effectiveness, enhancing decision-making processes for faster implementation, ensuring transparency, and developing skilled workforces that are capable of delivering towards nationally defined objectives.



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Oskar Gruber
Head of Management
Consulting Austria



Rihaan Samuel
Senior Consultant



Dinara Ibrayeva
Consultant



Oksana Syrotynska
Consultant

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Wolfgang Widmann
Senior Advisor ILF Austria



Dr. Andreas Linke
Senior Advisor ILF Germany



Yerzhan Yussupov
Head of ILF Uzbekistan



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